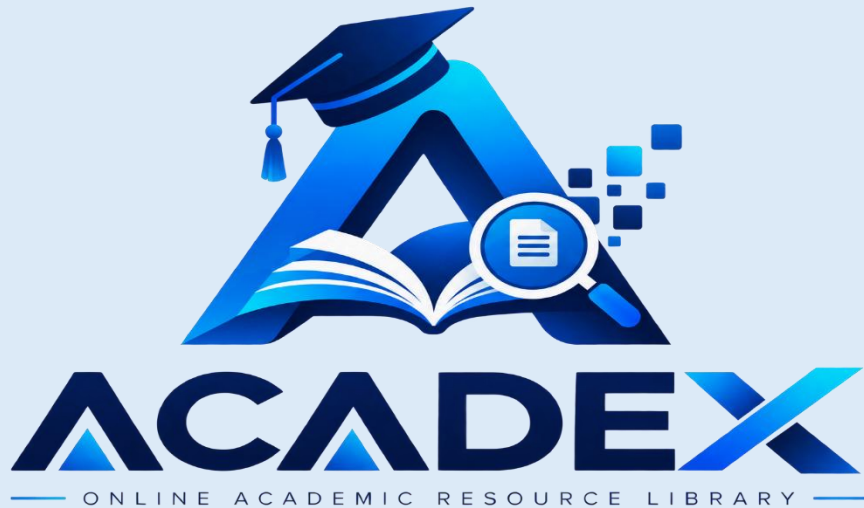




ACADEX

ONLINE ACADEMIC RESOURCE LIBRARY

PROJECT PROPOSAL

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IDENTIFICATION OF NEED

The availability of academic resources for studying and research is ever so limited and hard to get a hold of. This issue is especially prevalent on the Richfield Umhlanga campus. The available, recommended and most used academic resource allocation tool (Moodle) is an extremely limited web application.

Moodle is primarily used by lecturers to share academic resources for learning, revision and studying with students, which is heavily reliant on lecturers having to manually add academic resources themselves. Should they forget to do so, students are rendered helpless since they do not receive academic resources to use for studying and research purposes. The website is primarily used by the students to submit assignments, write formal assessments and have access to the aforementioned academic resources. It is important to note that sed resources are renewed every semester and past semester's resources are removed and rendered inaccessible to the students. Some learning material is not available because lecturers forget to add these resources which results in students having to look for other available online student libraries to try and access learning material. This can result in inefficient resource management, difficulty locating documents

However, upon research it is evident that for Richfield students it is quite difficult to find academic resources catered for the qualifications and courses offered by this institution in one singular website but are instead spread out amongst several various online student websites such as arXiv, ACM Digital Library, IEEE Explore, ScienceDirect, OpenLibrary and LimKey Nomad. This proves to be inefficient and time-consuming for students because they have to sieve through a plethora of websites to find the information they are looking for.

That is the problem Acadex will mitigate for Richfield students with the possibility of expanding beyond the confines of Richfield branching out nationally. Providing an online academic resource library that will serve Richfield students as an integrated resource for discovering, accessing, organizing and sharing academic resources. The system will provide a digital environment that will enable students to access academic content without being solely reliant on a physical library or using multiple platforms unrelated to academic content.

As previously stated, the project aims to develop a well-organized and user-friendly resource platform for students to easily access and engage with academic materials, inspired by academic resource sharing websites like Studocu. The primary goal of Acadex is thus to make academic resources more accessible, organized and discoverable via a single online platform.

PRELIMINARY INVESTIGATION

The purpose of this preliminary investigation is to examine the current libraries used by Richfield students and identify their strengths and weaknesses to build and manage a superior online student library for Richfield students and possibly more tertiary level students nationally. This investigation aims to identify existing challenges, determine the needs of the users and establish whether there is a requirement for a computerized information system. This investigation aims to determine whether or not the existing online libraries available to students are helpful or not and how they can be improved upon. This is further pushed by the increased dependency on AI tools that are crippling the basic comprehensive skills of learners which is a consequence of the lack of access to online libraries.

As it stands, students have to surf the internet thoroughly to find academic resources that they can use. They have to resort to being dependent on WhatsApp to receive academic resources from their lecturers and to be able to share academic resources amongst each other. Other student libraries make it a bit difficult to find resources as a consequence of offering academic resources for a wide selection of varied topics which trickles into the second problem. The inability to access past academic resources due to the library management style. Every semester the academic resources from the previous semester are removed and the material for the new upcoming semester is uploaded. The selection of available documents shared in libraries used by institutions such as Moodle is limited to what the lecturers deem to be sufficient.

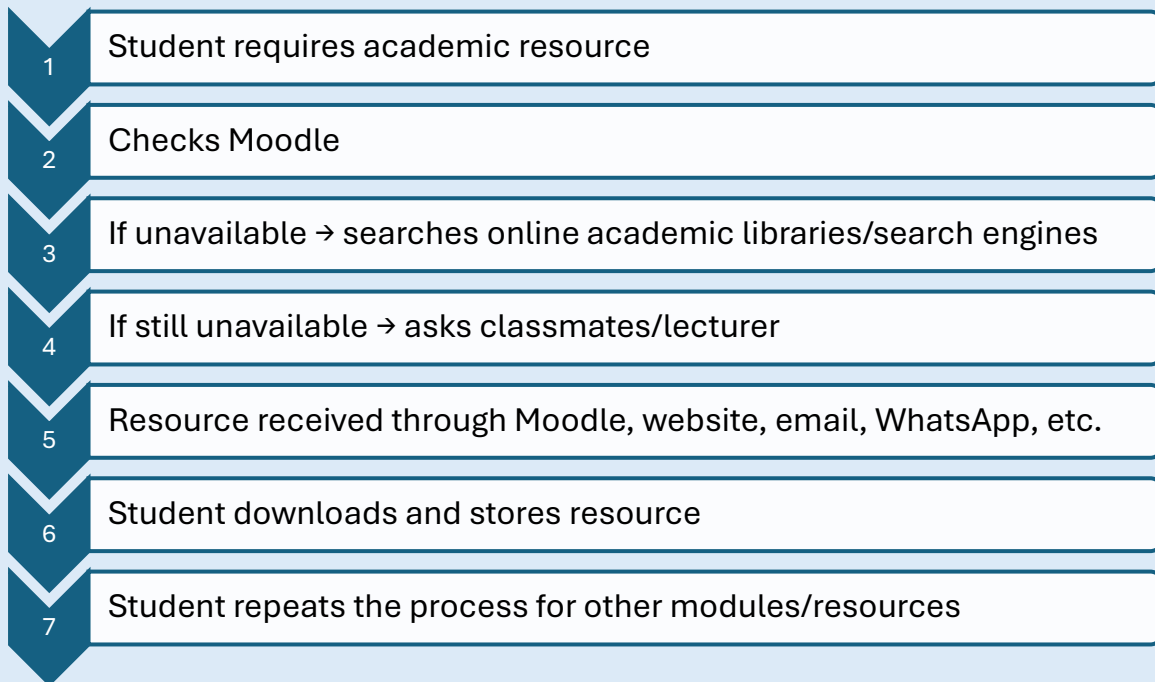
The selection of documents being limited to what the lecturers deem to be effective, disregards the different learning styles of students and the skill gap between lecturers and students. A more effective way would be to offer a selection of documents that the lecturers can add since they are granted administrative liberties as an institution and the students being able to add documents of their own findings because it is apparent that there is a never-ending supply of academic resources and none are better at finding them than the very students who use them. The ability to share sed resources amongst each other increases the variety of shared academic resources without flooding it with unrelated topics since the system would be specific to the Richfield campus and the qualification offered by Richfield. This would also combat the need to share resources repeatedly on WhatsApp group chats that can feel like an invasion of privacy due to the constant communication after campus learning hours. It would mitigate the need to contact a lecturer to remind them to share documents. The selection of academic resources being specific to the needs of Richfield students would narrow down the amount of Google searches a student would need to do since all the resources related to their modules would be available on this platform. Should sed student come across other learning material - while perusing through the internet – that they deem to be helpful they will have the ability to add it to the system. The addition of a chatbot that aids the

students with basic understanding, summaries and being a navigational assistant that helps direct a student towards the documents that they were searching for would be a great countermeasure for decreasing the dependency on artificial intelligence as a shortcut to finish assignments quickly but rather be used as a tool to help with learning and studying. The overall layout and design were also a point of contention. The lack of accessibility features that are available in these student libraries is a cause for concern. Most modern applications have the ability to change the display from light to dark mode, decrease or increase contrast, and text-to-speech features. It would be extremely wise to ensure that such features are implemented for the sole purpose of ensuring inclusivity and that the website is accessible to as many students as possible despite their impairments.

A questionnaire was distributed to students to determine how they currently access academic resources and what difficulties they experience. The questionnaire results indicated that 50 out of the 71 respondents were not satisfied with the selection of available e-books and online journals available on online libraries and when asked if they were able to find the academic sources you need for your classes 43 out of the 71 respondents were not able to find the resources they needed for class. This suggests that resource availability is an issue that should be considered when developing the proposed system. When asked about the addition of a chatbot onto the online library 65 out of the 71 respondents agreed, which suggested that the inclusion of an AI chatbot should be considered when development of the proposed system. The questionnaire results indicated that 43 out of 71 respondents experienced technical errors and/or slow loading pages while browsing which suggests that this is an issue that should be considered when designing the proposed system. An overwhelming majority of the respondents rated the overall design and layout of the website to be average with a small minority either leaning towards bad or good. Although aesthetics is subjective adhering to design principles and doing an adequate amount of research on UI/UX will ensure that this issue is considered and handled effectively in the design and development of the proposed system.

STAKEHOLDER	INTEREST IN SYSTEM	POSSIBLE REQUIREMENTS
Students	Provide resource Access resources	Upload, search, download, account
Lecturers	Provide resources	Upload and organize
Administrators	Manage system	User management, Security

CURRENT SYSTEM ANALYSIS



Problem and Errors Encountered

Difficulty locating resources which would be caused by a student may know that a particular document exists but not remembering whether it was uploaded to Moodle, sent through a WhatsApp group, stored on an online library, or shared by another student resulting in repeated searching across multiple platforms. Inaccessible resources since online resources occasionally contain expired links, unavailable files, restricted access, or documents that require registration or payment which would mean students wouldn't be able to access the actual document regardless of whether or not they find it. Duplicate resources result in the same document being uploaded or shared multiple times by different students and without a centralized system for identifying existing resources, students can accumulate duplicate files and waste time determining which version to use. Documents shared through messaging applications can become difficult to retrieve as conversations grow. Having to scroll through a large number of messages or ask a classmate to resend a document may prove to be problematic. File-management is another problem that arises because after downloading resources from different platforms, students may store them in different folders on their computers or phones and without consistent naming and categorization, locating a previously downloaded resource can become difficult.

In conclusion, the preliminary investigation established that the current online student libraries available to students present challenges relating to accessibility, search, record management, communication barriers and resource availability. The feedback collected from students indicates a requirement for a centralized method of managing, accessing and sharing academic resources. These findings provide a basis for further analysis of the proposed system and will be used to develop the system requirements and assess its feasibility.

FEASIBILITY

Acadex is proposed as a small-scale online academic resource library that will provide students with a centralized platform for discovering, accessing, organizing, and sharing academic resources. Although the initial system will be relatively small, the design will allow the system to be expanded with additional functionality in the future.

Technical feasibility

Considering the fact that the proposed system is to be a small-scale small scale online student library that has to be a scalable model. The proposed system can be developed using HTML, CSS and JavaScript for the frontend, making use of frameworks such as Bootstrap and Flask which will double as the frontend and backend framework since python will be used to connect the database to the system. MySQL will be used to handle database management because they are widely available and can operate on standard computer hardware. The CI/CD will be handled through the ACADEX GitHub repository. Security requirements can be met by ensuring that users are protected by using HTTPS, appropriate database access controls and avoiding the storage of passwords and sensitive credentials directly in the sources code. The development team is a group of third year undergrads, proficiency in programming languages is adequate for the scale of the proposed system. Network and internet are basic requirements because of the use of tools like GitHub. Fortunately, that is not an issue because the team has constant access to facilities that have the necessary network and internet specifications needed for the proposed system development process. The development team will require computers capable of running the development environment, Python, Flask, MySQL, a web browser, and a code editor such as Visual Studio Code. These requirements do not demand specialized hardware for the initial development of the system. The chatbot will be developed using Python. A basic chatbot can be implemented within the initial system, while more advanced chatbot functionality can be introduced in future versions (Hsieh, 2022).

Technical challenges include merge conflicts, stale local sync, infrequent commits, review bottlenecks, secret leakages, overprivileged access, network/offline dependency, inexperience, file storage, maintaining the system, database security and authentication (Perdana, 2022). Merge conflicts stem from two members editing the same file lines, requiring manual resolution, stale local sync which is a product of forgetting to pull latest changes which leads to divergence and broken builds. Large, rare dumps block visibility and compound merge pain. Loss of internet blocks remote sync and PR collaboration causing network/offline dependency. The inexperience of the developers is also a challenge that must be met with teamwork and collaboration. Authentication is an issue that relates to the possibility of an unauthorized user being given administrative control. But these are manageable within the proposed architecture and can be addressed through appropriate

development and testing practices. Therefore, Acadex is considered technically feasible for the proposed small-scale implementation (Perdana, 2022).

Operational Feasibility

The system should have a simple user interface that students can search for resources without requiring extensive training. Although it is important to note that the system doesn't have to be too simple because considering that the institution is a higher education institution, students have a basic level of computer literacy (Perdana, 2022). The proposed responsive interface will allow Acadex to be accessed from computers, tablets, and mobile devices. This increases the practicality of the system because students will not be restricted to using a specific type of device. The chatbot will further support operational usability by assisting students with navigating the system and understanding its functionality. This can reduce the difficulty that new users may experience when interacting with the platform. Administrators may require additional training to manage users, documents and system records. Administrative training is essential to ensure that there isn't mismanagement of student records, documents, academic resources and the ability to identify inappropriate content since much of the content will be user-generated the system will require appropriate reporting and administrative procedures. Users will therefore need to understand their responsibilities when uploading and interacting with resources. Considering the overwhelming negative response to currently available online student libraries, it is quite evident that the users will accept the new system (Perdana, 2022). The institution won't have to make any drastic changes considering the fact that this is a system primarily built for resources sharing and management. The only significant changes would be the use of the current system being used for assessments and assignment allocation, collection and grading moving forward. Overall, Acadex is operationally feasible because its proposed functionality corresponds with the needs of its intended users, while its administrative and reporting functions provide mechanisms for maintaining the platform (Perdana, 2022).

Economic Feasibility

The system is economically feasible if the expected benefits, such as reduced administrative effort and improved resource accessibility, justify the development, hosting and maintenance cost. Acadex is economically feasible because the proposed development technologies are simple to obtain because they have relatively low acquisition costs. Scalable architecture provides an economic advantage because of the future functionality that can be added in the future as the system grows. The project doesn't need to implement every possible feature during the initial development stage because resources can instead be allocated to the core functionality first, with additional features introduced as the system grows (Logmani, 2024). The development team can make use of existing computers for the development provided these computers meet the basic requirements for running the

tech stack/selected development tools which reduces the need to invest in specialized hardware. Since the development will be using open-source technology stack the initial licensing expenses will be mitigated. Potential costs may include hosting and storage, maintenance, domain registration, internet connectivity, security configuration, and future development; if the system grows significantly, increased storage, database capacity, and server resources may also generate additional costs and since the system can initially be deployed on relatively modest infrastructure because the expected number of users and resources will be limited during the initial implementation. As the number of users increases, additional infrastructure can be introduced gradually rather than requiring a large initial investment. For the proposed small-scale implementation, the expected technical infrastructure and software requirements can be accommodated without requiring substantial investment (Logmani, 2024).

PROJECT PLANNING

The identification of the activities that need to be completed for the successful development of the online student library make up the bulk of the planning of Acadex. The development of the system will be divided into different stages because Acadex is a small-scale system that is intended to be scalable. The planning process will focus on developing the main functions of the system first while still making sure that the system can accommodate additional functionality in the future. The division of the development of the system into stages is done to ensure that each part of the project can be worked on and tested before moving onto the next stage (Sheikhhoshkar, 2023).

The first stage of the project will involve identifying the requirements of the system and determining what problems the proposed system is expected to solve and involves the identification of the needs of students when it comes to accessing, finding and sharing academic resources. The team will also determine the main functions that Acadex needs to provide, such as user registration and login, searching for resources, uploading and downloading resources, saving resources, ratings, comments, reporting of inappropriate resources and administrative management. The chatbot will be considered during the planning stage so that it can be incorporated into the system rather than being treated as a separate feature after the main system has been completed (Sheikhhoshkar, 2023).

After the identification of the requirements is completed, the team will plan the structure and design of the system. This will involve deciding how the different parts of Acadex will work together and how information will be stored in the database. The design of the system will also consider how users will move between the different pages and how the interface will be structured so that students can easily find the functions they require (Sheikhhoshkar, 2023).

The development stage will then involve creating the frontend and backend of the system. HTML, CSS and JavaScript will be used to create the main interface, while Bootstrap will be used to assist with making the interface responsive across computers, tablets and mobile devices. Python and Flask will be used for the backend functionality and for connecting the system to the MySQL database. The development team will work on the different components of the system and integrate them together as development progresses. GitHub will be used to manage the source code and allow the team to keep track of changes made throughout the project (Sheikhhoshkar, 2023).

Testing being the most integral part of the project planning phase due to each major function needing to be checked before the system can be considered complete. The team will test functions such as registration, login, resource uploading, searching, downloading, saving, ratings, comments and reporting. The database will also be tested to ensure that information is stored correctly and that relationships between users and resources are maintained. Security will also be considered during testing by checking that unauthorized users cannot access restricted functions and that

sensitive information such as passwords and database credentials is handled appropriately (Sheikhhoshkar, 2023).

The chatbot will be developed and tested as part of the system so that it can assist students with navigating Acadex and understanding how to use the platform. The initial chatbot will focus on providing basic assistance because the system is intended to be a small-scale implementation as such additional chatbot functionality will be introduced in the future as the system is expanded (Sheikhhoshkar, 2023).

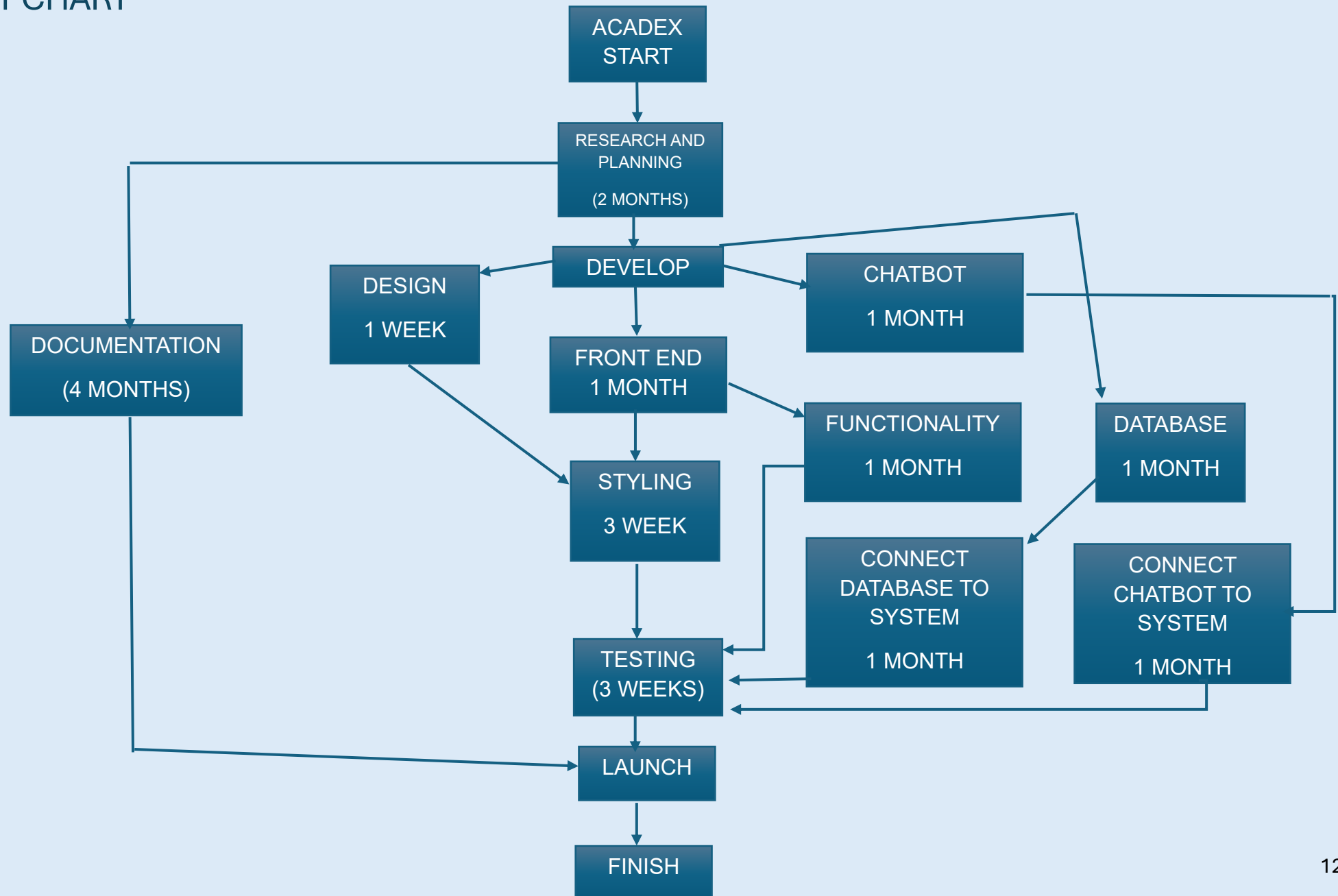
The final stage of the project involves the integration of all the developed components and performing overall system testing so any errors or problems identified during testing will be corrected before the system is presented as a completed project. And to determine whether or not the main objectives of Acadex have been achieved the team will review the system against the original requirements. Documentation will be completed alongside the development process so that the final report accurately represents the system that was designed and implemented (Sheikhhoshkar, 2023).

The project will therefore follow a structured development process consisting of requirements identification, system planning and design, database design, frontend and backend development, chatbot development, testing, integration and final evaluation. This approach will allow the team to manage the development of Acadex in stages while ensuring that the main requirements are addressed and that the system remains capable of being expanded with additional functionality in the future (Sheikhhoshkar, 2023).

GANTT CHART

	TASK	ASSIGNED TO	START	END	%	AUG	SEP	OCT	NOV
	ACADEX		8/4/26	9/11/26	25%				
1	DOCUMENTATION	A.RADEBE	8/4/26	9/11/26	40%				
2	FRONTEND – HTML	N.MTHETHWA	15/8/26	9/10/26	60%				
3	FRONTEND – CSS	J.GASA	21/8/26	15/10/26	10%				
4	FRONTEND – JAVASCRIPT	S.MFEKA	9/9/26	16/10/26	10%				
5	BACKEND	D.XHAKAZA	9/9/26	13/10/26	10%				
6	DATABASE	N.MAHLOBO	15/8/26	9/10/26	40%				
7	DEPLOYMENT	T.TSHEBE	1/9/26	16/10/26	10%				

PERT CHART



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

The definition of the functional and non-functional requirements of Acadex is the purpose of this Software Requirements Specification.

Acadex is an online academic resource library proposed by that will serve students as an integrated resource for discovering, accessing, organizing and sharing academic resources. The system will provide a digital environment that will enable students to access academic content without being solely reliant on a physical library or using multiple platforms unrelated to academic content. Acadex is proposed as a small-scale system that is capable of being expanded should the number of users, resources and system functions increase. The system is designed to provide students with an alternative to relying solely on physical libraries or multiple unrelated online platforms when searching for academic resources (Zahran, 2025).

Acadex will be a web-based client-server application academic resource platform that allows registered students to upload, search for, access, download, save, rate, and comment on academic resources along with resources being organized according to subjects, courses, and categories to make them easier to discover. The system will also provide mechanisms for users to report inappropriate, inaccurate, or unsuitable content. The chatbot will operate as an additional component of the application and will communicate with the system to provide users with navigation and assistance. Administrators will have access to management functionality that allows them to manage users and resources and maintain the quality of the platform. A chatbot will be incorporated into the system to assist users with navigation and provide basic assistance when using Acadex (Zahran, 2025). The system will use HTML, CSS, JavaScript, and Bootstrap for the user interface, while Python and Flask will provide the application and server-side functionality. MySQL will be used to manage the system's data. GitHub will be used for source-code management and CI/CD activities. With regards to usage of the system, the primary users of Acadex will be students who require access to academic resources. Users will be able to register accounts and interact with resources available on the platform. They will interact with the system through a web browser, while the Flask application will process requests and communicate with the MySQL database. Administrators will make up a second user group, and they will be responsible for managing users, resources, reports, and other administrative aspects of the system (Zahran, 2025).

The functional requirements describe the operations and services that Acadex must provide for its users. The system will allow new users to create an Acadex account. The registration process will collect the information required to create and identify a user account. The system shall The system will prevent duplicate accounts where the same unique identifying information has already been registered, allow registered users to log into their accounts, authenticate submitted login credentials before granting access to protected functionality and restrict authenticated functions,

such as uploading resources and saving resources, to authorized users because it validate the submitted information before creating the account. User passwords will not be stored as plain text therefore passwords will be securely hashed before being stored in the database and only authenticated users will be able to access their account and manage relevant account information, and the system will maintain information associated with each registered user, including their account credentials and relevant activity within Acadex. User access will be controlled according to the user's role, such as student or administrator (Zahran, 2025).

The system will allow registered users to upload academic resources. When uploading a resource, the user will provide relevant information such as the resource title, subject, course, and category. The system should associate the uploaded resources with the account of the user who uploaded them and it should also validate uploaded files and information before storing them. The system shall organize academic resources according to appropriate subjects, courses, and categories. Administrators will be able to manage resource classifications where necessary. Users will be able to search using relevant keywords and the system shall support searching and filtering based on information such as subjects, courses, and categories. The system will allow users to view available academic resources where the resource format supports online viewing. The system will allow users to download available academic resources where downloading is permitted. The system will allow authenticated users to save resources for easier access at a later time and the saved resources shall be associated with the user's account and will be retrievable through the user's account interface (Zahran, 2025).

The system should allow users to rate academic resources and display rating information to assist users in evaluating resources. The system will allow users to submit comments on academic resources. Comments shall be associated with the relevant resource and the user who submitted them. The system will allow users to report academic resources that they believe are inappropriate, inaccurate, unsuitable, or otherwise problematic and administrators shall be able to review submitted reports and take appropriate administrative action. The system shall provide administrators with administrative functionality for managing Acadex. Administrators shall be able to manage registered users and academic resources. An integrated chatbot to assist users with navigating Acadex also needs to be provided by the system. The chatbot should provide basic guidance relating to the use of the system, such as locating resources, understanding available functions, and navigating different sections of the platform. It also important that the chatbot is designed so that additional capabilities can be incorporated in future versions of Acadex (Zahran, 2025).

The non-functional requirements define the expected quality, performance, security, and operational characteristics of Acadex. Acadex should provide reasonable response times for common operations such as logging in, searching for resources, viewing resources, and accessing saved resources. Database queries should be designed efficiently to prevent unnecessary delays as the number of resources and users increases. The system should protect user information and system data from unauthorized access and communication between users, and the application should use HTTPS to protect information transmitted over the network (Saroja, 2023).

The database access shall be controlled using appropriate authentication and authorization mechanisms. Sensitive credentials, database passwords, secret keys, and other confidential configuration information shall not be stored directly within the application's source code. The system should operate consistently and correctly when users perform normal operations. The application should handle invalid input and unexpected requests without exposing sensitive system information or causing unnecessary system failure. Acadex should provide a simple and understandable interface that allows students to navigate the system with minimal difficulty, common functions such as searching, uploading, downloading, saving, and accessing resources should be clearly identifiable. The system will be designed as a scalable model. Although the initial implementation will be small-scale, the architecture should allow additional users, resources, and functionality to be incorporated without requiring the entire system to be redesigned (Saroja, 2023). The system should be structured in a manner that allows developers to identify, modify, test, and maintain individual components without unnecessarily affecting other components. The use of Python, Flask, modular application components, Git, and GitHub will support collaborative development and future maintenance and the source code should be documented sufficiently to allow team members and future developers to understand the implementation. Acadex shall be accessible through modern web browsers and the responsive design must allow the application to operate across desktop, tablet, and mobile environments. Acadex should be available to authorized users whenever the application and hosting environment are operational. The system shall maintain the accuracy and consistency of stored information. The relationships between users, resources, ratings, comments, saved resources, and reports shall be appropriately maintained within the database. Database constraints and validation mechanisms should be used to reduce invalid or inconsistent data (Saroja, 2023).

The user interface must provide users with access to the major functionality of Acadex.

The interface should include appropriate pages or components for registration, login, resource discovery, resource viewing, resource uploading, saved resources, user interaction, and chatbot assistance.

Administrative users will have access to additional interfaces for managing users, resources, and reports.

The Acadex application will interact with:

- A web browser through HTML, CSS, JavaScript, and Bootstrap.
- The Flask application framework through Python.
- The MySQL database through an appropriate Python database connector.
- GitHub for source-code management and CI/CD processes.

The application will communicate with the MySQL database to store and retrieve system information.

The database will contain information relating to users, academic resources, subjects, courses, categories, ratings, comments, saved resources, and content reports as such access to the database will be restricted to authorized application processes and users.

The initial version of Acadex will be developed as a small-scale academic resource platform. The proposed technology stack consists primarily of HTML, CSS, JavaScript, Bootstrap, Python, Flask, and MySQL. And the initial system may provide a limited set of chatbot capabilities compared with advanced commercial artificial-intelligence systems, but the chatbot can be expanded in later versions as additional requirements and resources become available. The system will also depend on an internet connection for users to access the online platform.

The system assumes that users have access to a compatible device and an internet connection. It is assumed that users will provide valid information when registering and uploading resources. The system depends on the availability and correct configuration of the Flask/Python application environment, MySQL database, web server, and hosting environment. The system also depends on GitHub for collaborative source-code management and the proposed CI/CD workflow.

The accuracy and usefulness of resources uploaded by users cannot be guaranteed automatically. The rating, commenting, reporting, and administrative functions will therefore assist in identifying potentially unsuitable content.

Future versions could expand the chatbot's capabilities and introduce more advanced features. The initial implementation will therefore focus on establishing the core resource-library functionality while maintaining an architecture capable of supporting future expansion.

In conclusion the Software Requirements Specification defines the requirements for the development of Acadex as a small-scale, scalable online academic resource library. The system will provide students with a centralized platform for discovering, accessing, organizing, and sharing academic resources.

The combination of HTML, CSS, JavaScript, Bootstrap, Python, Flask, and MySQL provides the proposed technical foundation for the system.

DATA MODELS

The Acadex data model defines the manner in which information in the online student library will be structured, stored, and related. Considering how Acadex is going to use MySQL as its database management system, the data model will be implemented using a relational database structure which will support Acadex's core functions which are user registration, resource uploading, searching, downloading, saving resources, ratings, comments, content reporting, and administrative management (Mukhlis, 2023). The structure also allows the system to be expanded as additional functionality is introduced.

The main entities of the proposed Acadex database are **(User, Resource, Subject, Course, Category, Rating, Comment, Saved Resource, Report, and Chatbot Interaction)**. These entities will be connected through relationships that represent how users and resources interact within the system.

The **User** entity stores information about students and administrators who use Acadex.

The user_id will serve as the primary key. The email address should be unique to prevent duplicate accounts.

ATTRIBUTE	DATA TYPE	DESCRIPTION
user_id	INT	Primary key that uniquely identifies a user
first_name	VARCHAR	User's first name
last_name	VARCHAR	User's surname
email	VARCHAR	User's unique email address
password_hash	VARCHAR	Securely hashed user password
role	VARCHAR	Identifies whether the user is a student or administrator
date_registered	DATETIME	Date and time the account was created

The **Resource** entity represents academic documents and other learning materials uploaded to Acadex.

The user_id, subject_id, course_id, and category_id will act as foreign keys connecting the resource to the relevant entities

ATTRIBUTE	DATA TYPE	DESCRIPTION
resource_id	INT	Primary key identifying the resource
title	VARCHAR	Name of the academic resource
description	TEXT	Description of the resource
file_name	VARCHAR	Name of the uploaded file
file_path	VARCHAR	Location of the stored resource
user_id	INT	Foreign key identifying the uploader
subject_id	INT	Foreign key identifying the subject
course_id	INT	Foreign key identifying the course
category_id	INT	Foreign key identifying the category
upload_date	DATETIME	Date and time the resource was uploaded
status	VARCHAR	Indicates the current resource status

The **Subject** entity stores the academic subjects under which resources can be organized.

For example, subjects could include Information Technology, Mathematics, Computer Science, or Accounting.

ATTRIBUTE	DATA TYPE	DESCRIPTION
subject_id	INT	Primary key
subject_name	VARCHAR	Name of the subject
description	TEXT	Description of the subject

The **Course** entity stores information about academic courses or modules.

A subject can contain multiple courses. For example, an academic subject or department may contain several courses or modules.

ATTRIBUTE	DATA TYPE	DESCRIPTION
course_id	INT	Primary key
course_name	VARCHAR	Name of the course
course_code	VARCHAR	Unique course or module code
Subject_id	INT	Foreign key identifying the related subject

The **Category** entity allows resources to be grouped according to their type.

Possible categories could include lecture notes, assignments, past examination papers, textbooks, study guides, and research materials.

ATTRIBUTE	DATA TYPE	DESCRIPTION
category_id	INT	Primary key
category_name	VARCHAR	Name of the category
description	TEXT	Description of the category

The **Rating** entity stores ratings submitted by users for academic resources.

A user should generally be restricted from repeatedly submitting multiple ratings for the same resource. A database constraint involving user_id and resource_id can be used to support this rule.

ATTRIBUTE	DATA TYPE	DESCRIPTION
rating_id	INT	Primary key
user_id	INT	Foreign key identifying the user
resource_id	INT	Foreign key identifying the resource
rating_value	INT	Rating given to the resource
rating_date	DATETIME	Date and time of the rating

The **Comment** entity stores comments submitted by users about academic resources.

The relationship allows multiple users to comment on the same resource while allowing each comment to be associated with its author.

ATTRIBUTE	DATA TYPE	DESCRIPTION
comment_id	INT	Primary key
user_id	INT	Foreign key identifying the user
resource_id	INT	Foreign key identifying the resource
comment_text	TEXT	Content of the comment
comment_date	DATETIME	Date and time the comment was created

The **Saved Resource** entity records resources that users have saved for future access.

A user can save multiple resources, and a resource can be saved by multiple users. The Saved Resource entity therefore acts as an associative entity between users and resources.

ATTRIBUTE	DATA TYPE	DESCRIPTION
saved_id	INT	Primary key
user_id	INT	Foreign key identifying the user
resource_id	INT	Foreign key identifying the saved resource
saved_date	DATETIME	Date and time the resource was saved
saved_id	INT	Primary key

The **Report** entity stores reports submitted by users regarding potentially inappropriate, inaccurate, or unsuitable resources.

The report status could indicate whether a report is **pending, under review, resolved, or dismissed**.

ATTRIBUTE	DATA TYPE	DESCRIPTION
report_id	INT	Primary key
user_id	INT	Foreign key identifying the reporting user
resource_id	INT	Foreign key identifying the reported resource
reason	VARCHAR/TEXT	Reason for the report
report_date	DATETIME	Date and time the report was submitted
status	VARCHAR	Current status of the report
admin_id	INT	Administrator responsible for reviewing the report

The **Chatbot Interaction** entity can be used to store interactions between users and the Acadex chatbot.

For the initial small-scale implementation, storing chatbot interactions may be optional. If implemented, the entity provides a structure for maintaining interaction history and potentially improving future chatbot functionality.

ATTRIBUTE	DATA TYPE	DESCRIPTION
interaction_id	INT	Primary key
user_id	INT	Foreign key identifying the user
user_message	TEXT	Message submitted by the user
chatbot_response	TEXT	Response generated by the chatbot
interaction_date	DATETIME	Date and time of the interaction

Entity Relationships

The main relationships within the Acadex database are as follows:

User – Resource

A **User can upload many Resources**, while each Resource is uploaded by one User.

Relationship: User (1) → (Many) Resource

Subject – Course

A **Subject can contain multiple Courses**, while each Course belongs to a particular Subject.

Relationship: Subject (1) → (Many) Course

Course – Resource

A **Course can contain many Resources**, while each Resource can be associated with a Course.

Relationship: Course (1) → (Many) Resource

Category – Resource

A **Category can contain many Resources**, while each Resource can belong to a category.

Relationship: Category (1) → (Many) Resource

User – Rating – Resource

A User can rate multiple Resources, and a Resource can receive ratings from multiple Users. The Rating entity resolves this many-to-many relationship.

Relationship: User (1) → (Many) Rating / Resource (1) → (Many) Rating

User – Comment – Resource

A User can submit multiple comments, and a Resource can have multiple comments. The Comment entity connects users and resources.

Relationship: User (1) → (Many) Comment / Resource (1) → (Many) Comment

User – Saved Resource – Resource

A User can save many Resources, while a Resource can be saved by many Users. The Saved Resource entity resolves this many-to-many relationship.

Relationship: User (1) → (Many) Saved Resource / Resource (1) → (Many) Saved Resource

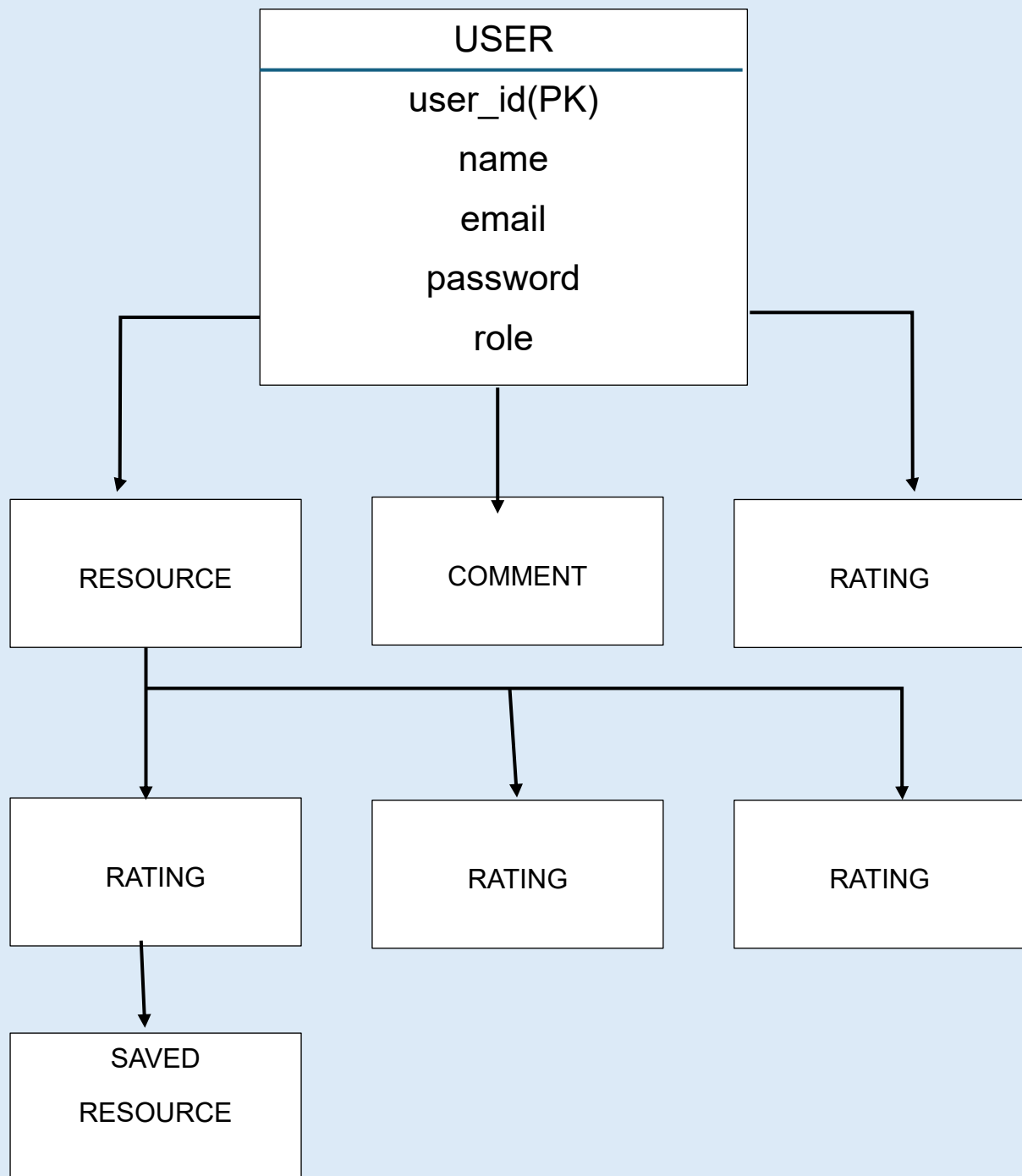
User – Report – Resource

A User can submit multiple reports, while a Resource can be reported multiple times. The Report entity records these interactions.

Relationship: User (1) → (Many) Report / Resource (1) → (Many) Report

(Mukhlis, 2023)

Simplified Entity Relationship Model



USER -----REPORT-----RESOURCE

USER -----CHATBOT INTERACTION

The database will use **primary keys** to uniquely identify records and **foreign keys** to maintain relationships between entities (Mukhlis, 2023).

Input validation will be implemented to reduce invalid data entering the database. e.g. (email addresses should follow an appropriate format). Database constraints should also be used where and when it is appropriate. e.g. (the user's email address can be defined as unique, while combinations such as user_id and resource_id can be restricted in tables such as ratings and saved resources to prevent unnecessary duplicate records). Sensitive user information will be protected, and passwords will be stored as secure hashes rather than plaintext values, and database credentials will not be embedded directly within the application's source code (Mukhlis, 2023).

The proposed relational data model is designed to support the initial small-scale implementation while providing a foundation for future expansion. The additional entities that can be introduced without changing the existing structure for future versions could introduce entities for notifications, resource versions, user profiles, universities, departments, resource verification, search history, or advanced chatbot functionality. The separation of users, resources, academic classifications, user interactions, and administrative reports also makes the database easier to maintain and extend as Acadex grows (Mukhlis, 2023).

The proposed Acadex data model provides a structured relational foundation for managing the information required by the online student library. The model separates major system components into related entities while using primary and foreign keys to maintain relationships and data integrity (Mukhlis, 2023).

The model supports the core functionality of Acadex, including resource management, searching, user interaction, saving resources, content reporting, administration, and chatbot assistance. Its relational structure is appropriate for implementation using MySQL and provides a scalable foundation for the future development of the system (Mukhlis, 2023).

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